IBM Capstone Project

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IBM Developer

SKILLS NETWORK

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- Three Dashboards that represents technology worked with and desired and demographics of the respondents.

- Most Used -
  - Language: ‘JavaScript’
  - Database: ‘MySQL’
  - Web-frame: ‘jQuery’
  - Platform: ‘Windows’

EXECUTIVE SUMMARY

- Most Desired-
  - Language: ‘JavaScript’
  - Database: ‘PostgreSQL’
  - Web-frame: ‘React.js’
  - Platform: ‘Linux’

- Respondents-
  - By gender: 93.50% are Male
  - By countries: USA
  - By Education: Bachelor’s Degree
  - By age: 28

INTRODUCTION

- Which is the most used & desired language?
- Which is the most used & desired database?
- Which is the most used & desired web-frame?
- Which is the most used & desired platform?
- What gender & age most respondents are?
- Which country most respondents belong from?
- What formal education do most respondents have?

METHODOLOGY

Dataset source

https://www.kaggle.com/kyanyoga/sample-sales-data

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RESULTS

As for ‘Current Technology Usage’ dashboard:

1. The Bar Chart represents that 'JavaScript' ranks first and 'C++' ranks last as most used language.
2. The Column Chart shows that the most used database as 'MySQL' and the least used in top 10 as 'Firebase’.
3. The Word Cloud Chart shows that 'Windows' is most used and platforms other than major ones are least used platforms.
4. The Hierarchy Bubble Chart depicts that the 'jQuery' is the most and 'Laravel' is the least worked web-frame in top 10 used web-frame.

RESULTS

As for ‘Future Technology Trend’ dashboard:

1. The Bar Chart represents that 'JavaScript' ranks first and ‘Kotlin' ranks last as most desired language.
2. The Column Chart shows that the most used database as 'PostgreSQL' and the least used in top 10 as 'DynamoDB’.
3. The Tree Map Chart shows that ‘Linux' is most used and platforms other than major ones are least used platforms.
4. The Hierarchy Bubble Chart depicts that the ‘React.js' is the most and 'Ruby on Rails' is the least worked web-frame in top 10 used web-frame.

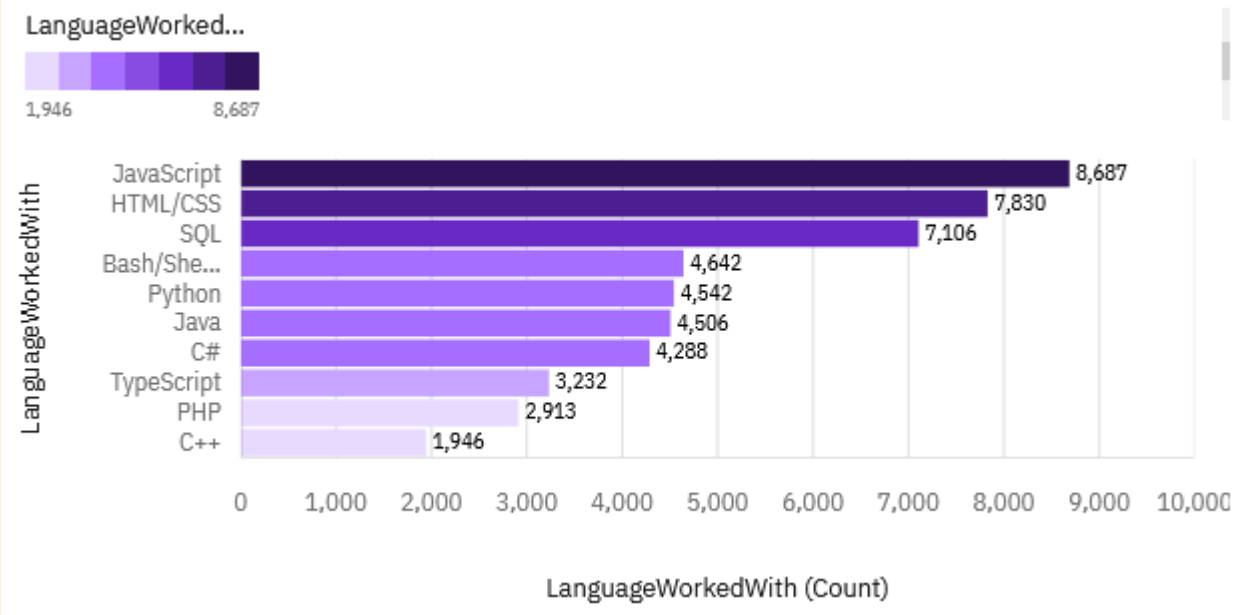
RESULTS

As for ‘Demographics’ dashboard:

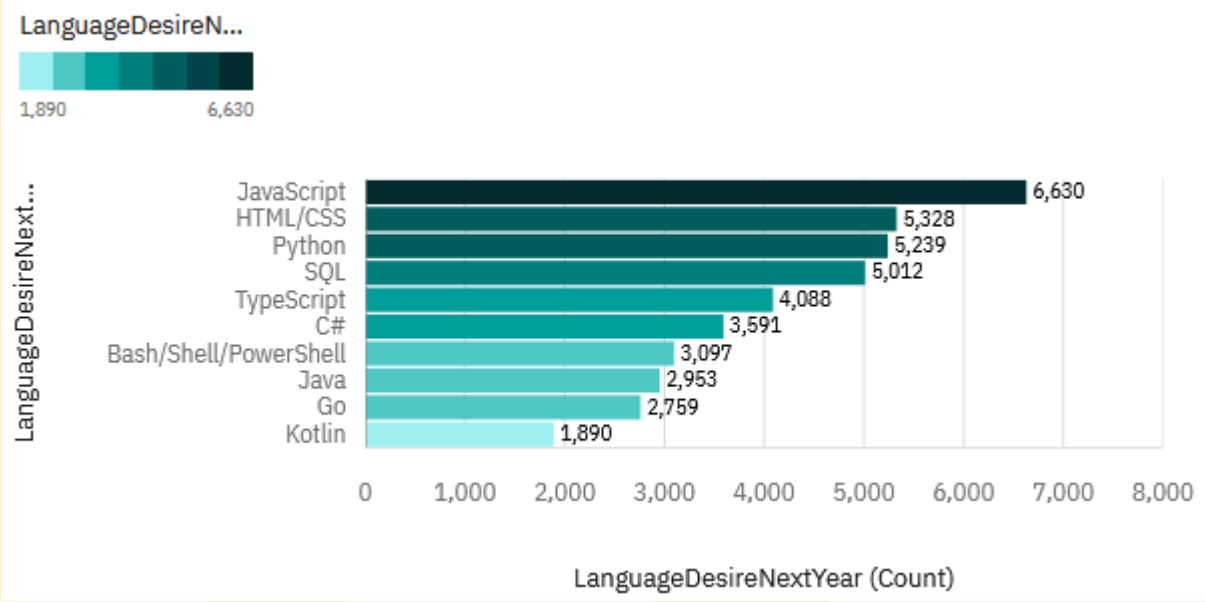
1. The Pie Chart represents that most respondents are ‘Male’ summing up to 93.50%.
2. The World Map Chart shows that the majority respondents are from USA.
3. The Line Chart shows that most are of aged 28 and only one is of 99 year.
4. The Stacked Bar Chart depicts that most have done Bachelor’s Degree for their formal education.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

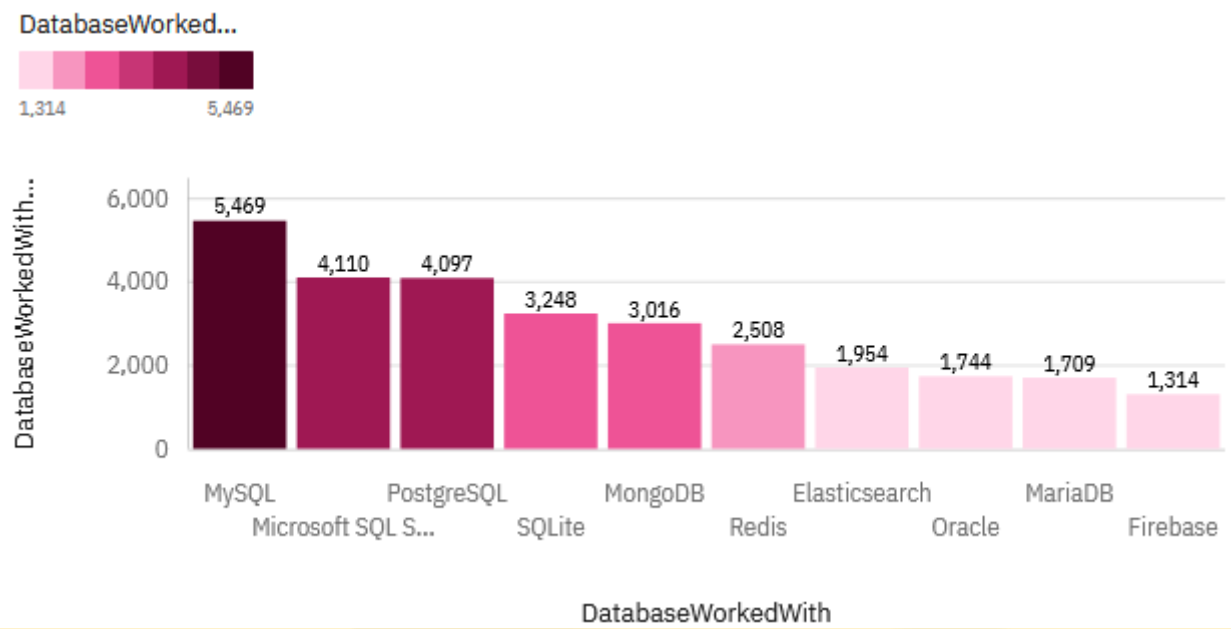
- JavaScript & HTML/CSS are most worked with and desired language
- SQL is third most used language but Python is the third most desired
- C++ & Kotlin are last in top 10 list as most used and most desired language respectfully.

Implications

- JavaScript is the most in demand language
- HTML/CSS are second in demand language
- Go & Kotlin aren't top 10 languages worked with and are in top 10 languages desired.

DATABASE TRENDS

Current Year



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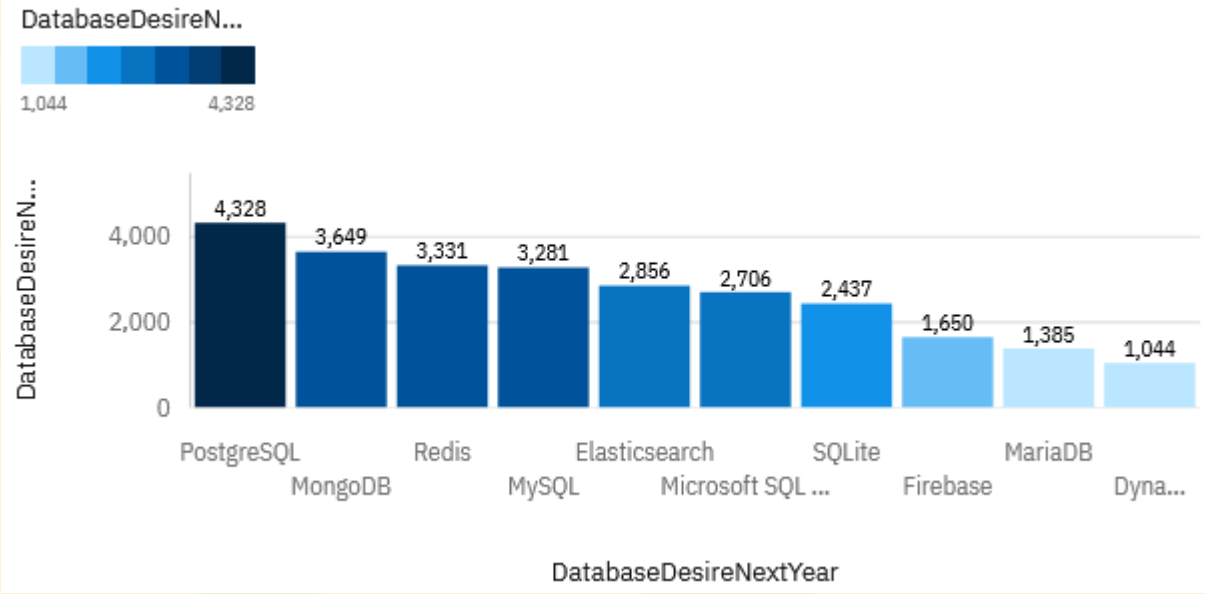


DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

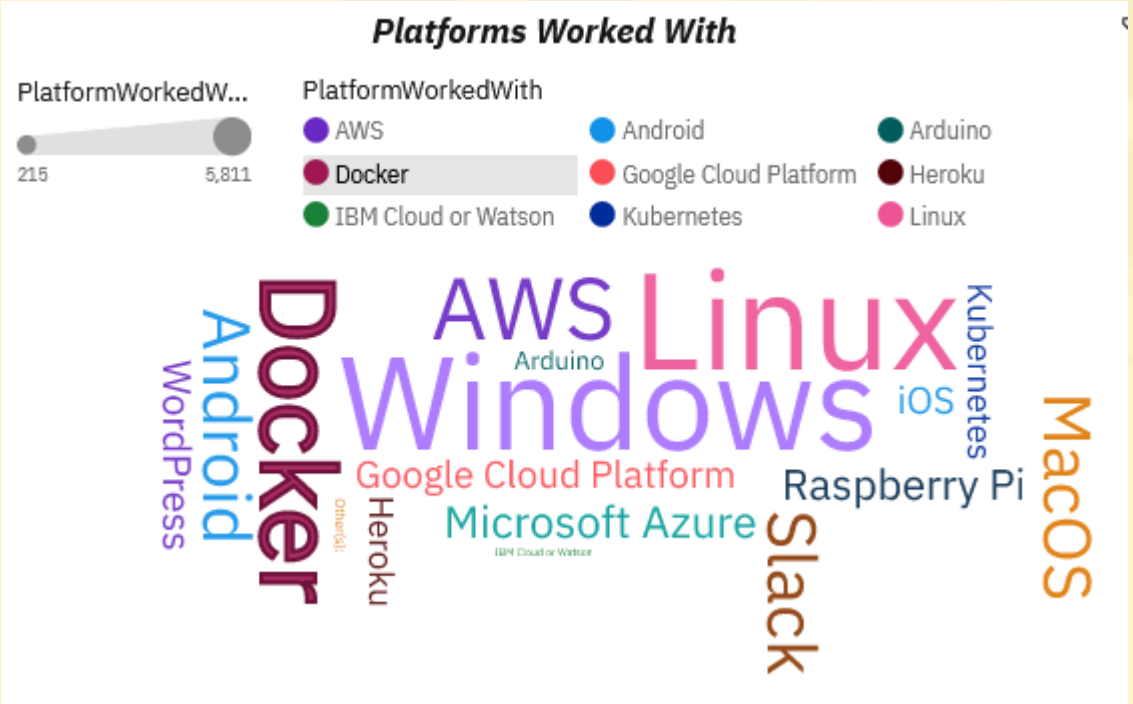
- MySQL is most worked with database
- Microsoft SQL Server & PostgreSQL are the runner ups in top 10 list

Implications

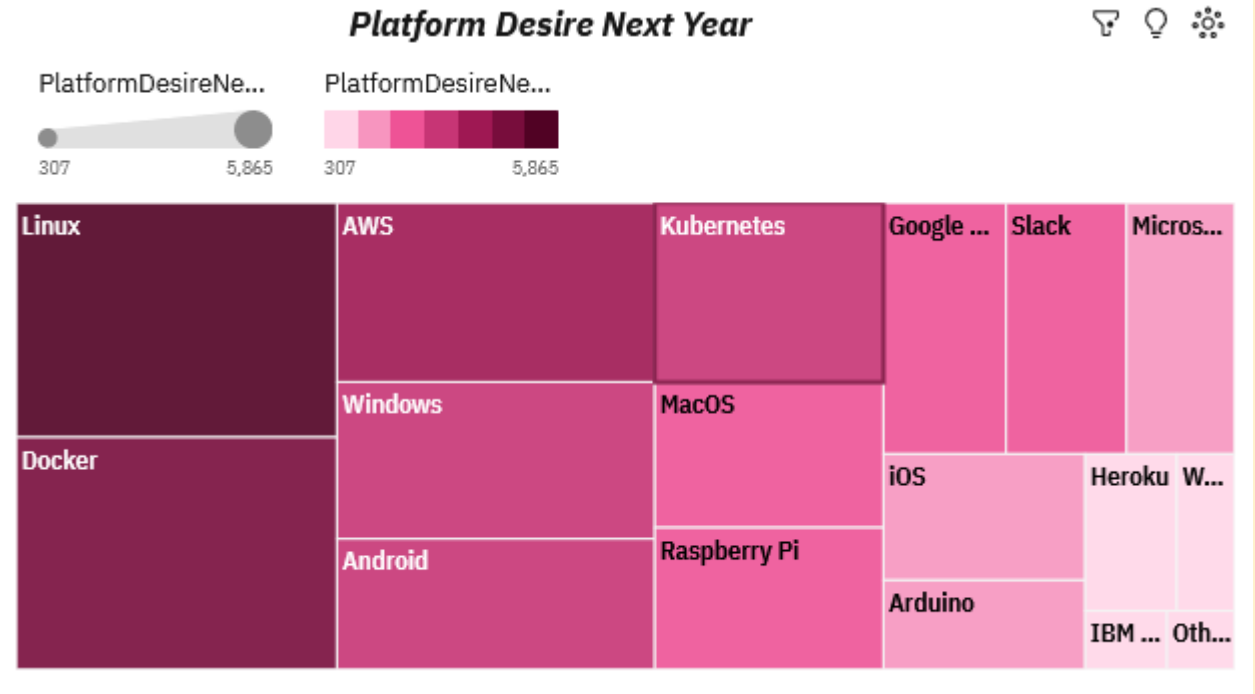
- PostgreSQL is the most desired database
- DynamoDB isn't in top 10 list of database worked with but is surely in top 10 list of desired database.

PLATFORM TRENDS

Current Year



Next Year



PLATFORMS TRENDS - FINDINGS & IMPLICATIONS

Findings

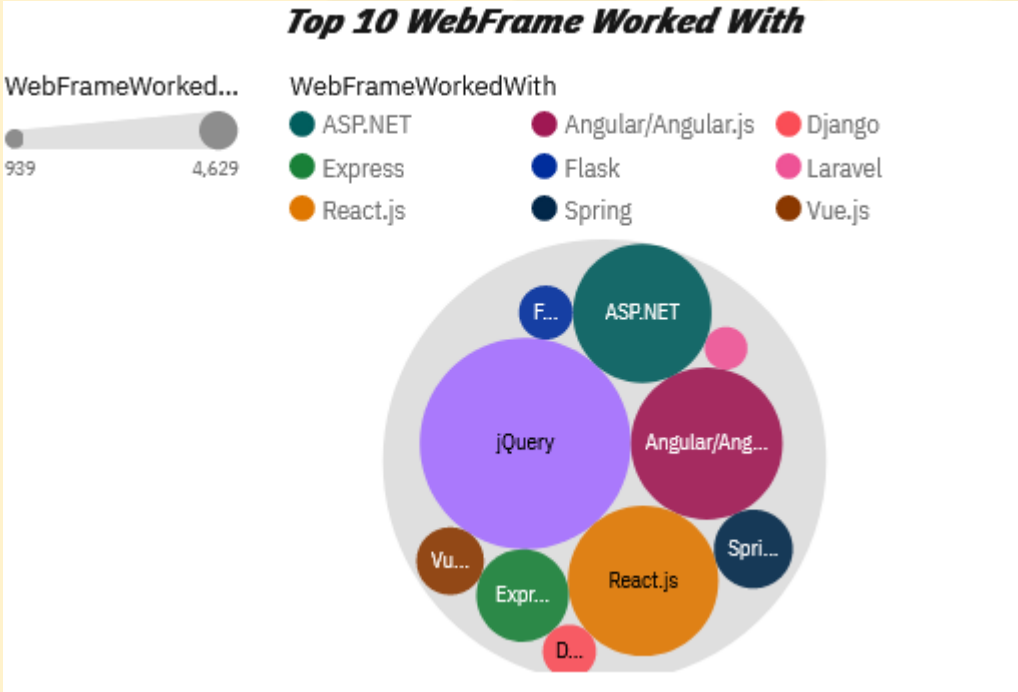
- Windows and Linux are the most used platform
- IBM Cloud or Watson is the least used platform

Implications

- Linux and Docker is the most desired platform
- IBM Cloud or Watson remains to be the least desired platform

WEB-FRAME TRENDS

Current Year



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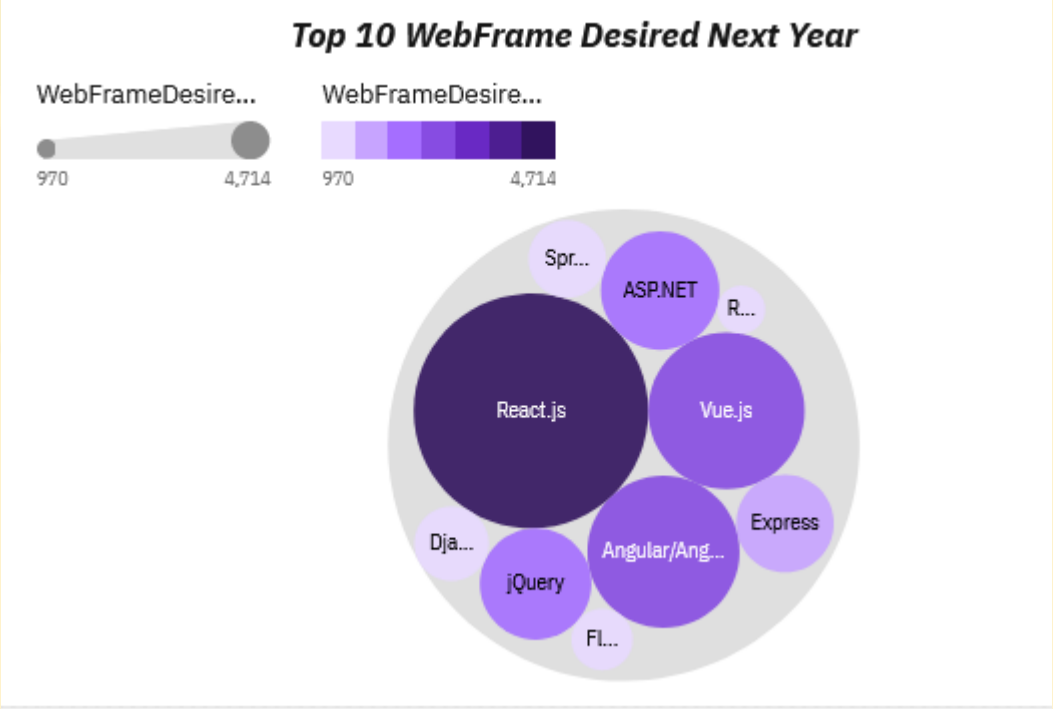


WEB-FRAME TRENDS - FINDINGS & IMPLICATIONS

Findings

- ‘j.Query’ and ‘React.js’ are the most used web-frame
- ‘Lavarel’ is the least used in top 10 list

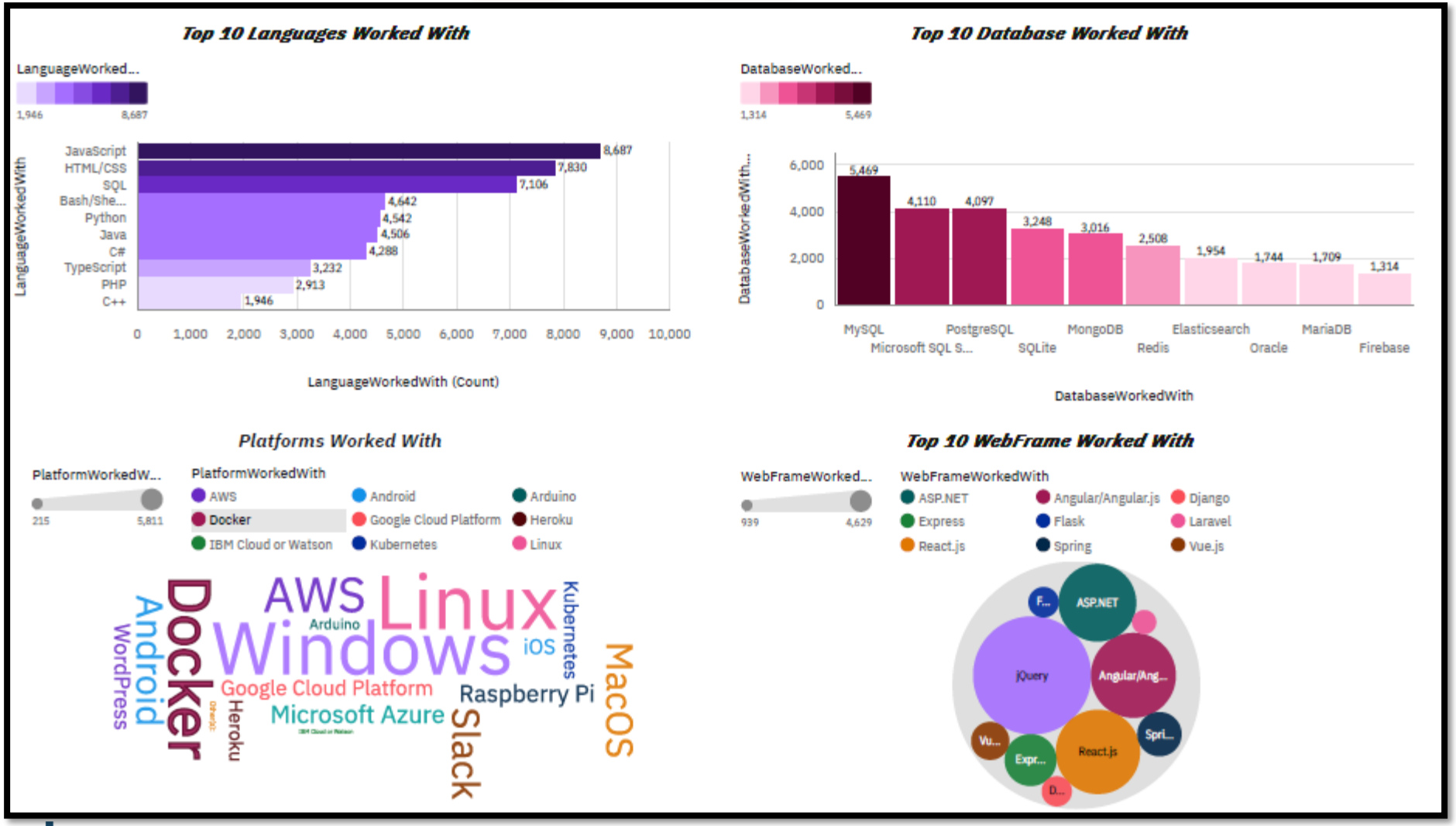
Implications

- ‘React.js’ is most desired to work with database
- ‘Ruby on Rails’ stands on 10th position in top desired web-frame

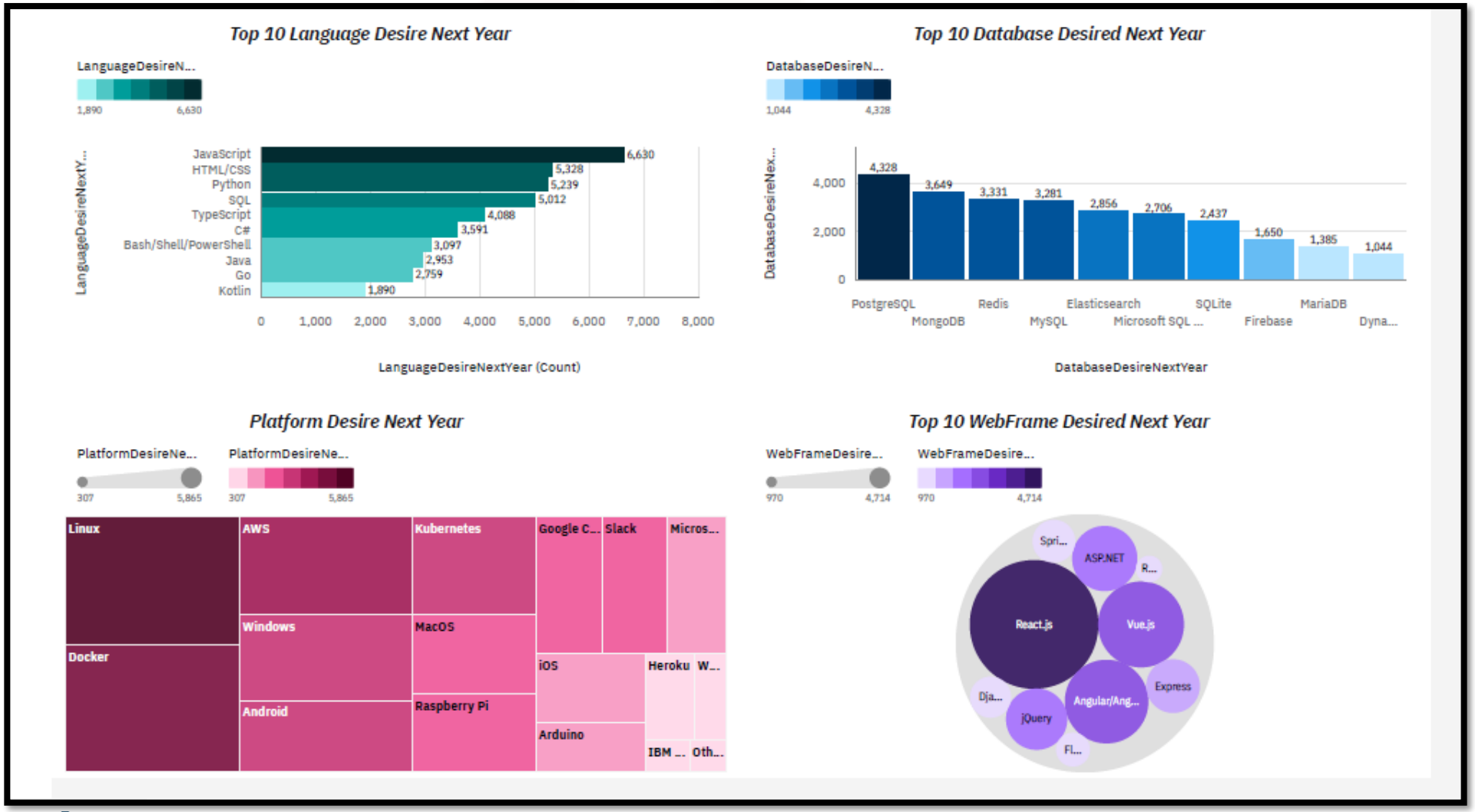
DASHBOARD

https://github.com/dprachi75/Cognos-Dashboard.git

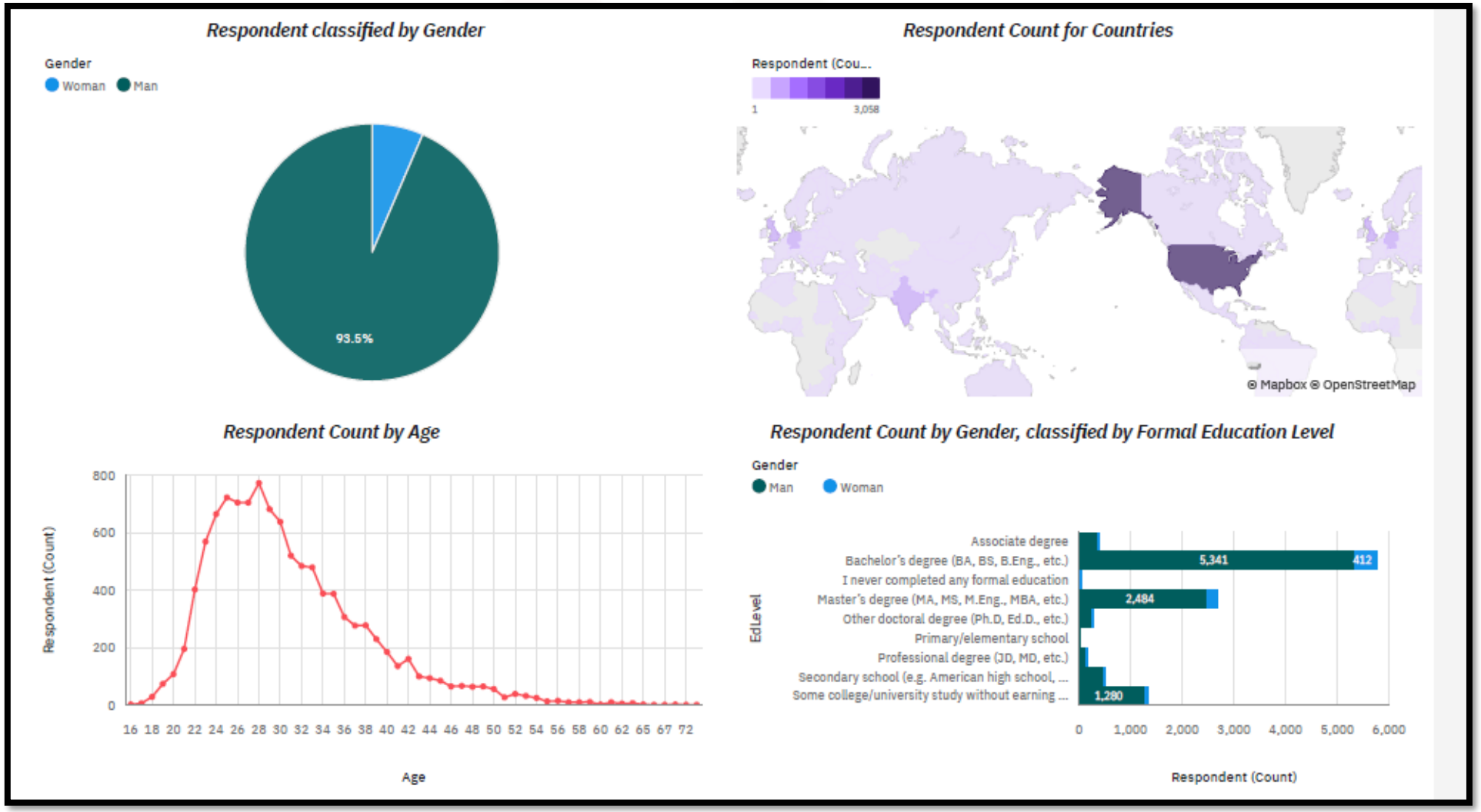
‘Current Technology Used’ Tab



‘Future Technology Trend’ Tab



‘Demographics’ Tab



DISCUSSION

OVERALL FINDINGS & IMPLICATIONS

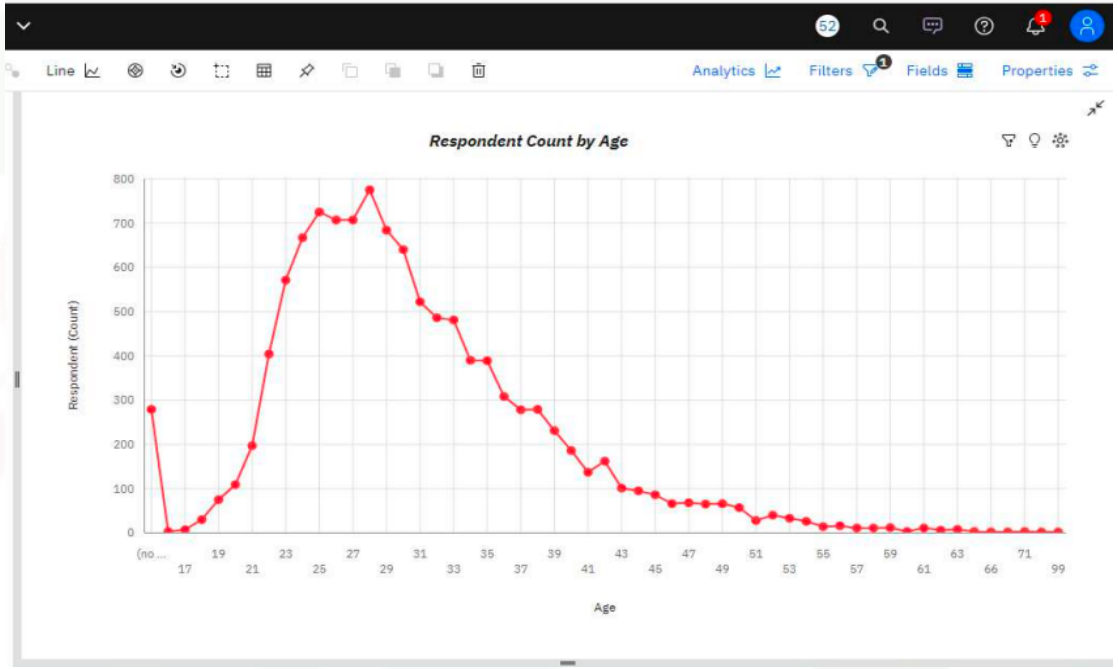
Male respondents aged 28, from USA holding Bachelor's degree are working with JavaScript, MySQL, Windows & jQuery.

Male respondents aged 28, from USA holding Bachelor's degree are willing to work with PostgreSQL, Linux & React.js.

CONCLUSION

- JavaScript is the most in demand language
- Windows is most used but Linux is in demand
- Most work on MySQL and are willing to work on PostgreSQL
- Many work on jQuery and wish to work on React.js

APPENDIX


Line
Analytics
Filters
Fields
Properties
Respondent Count by Age
Respondent (Count)
800
700
600
500
400
300
200
100
0
Age

JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.